# QUIZ: Transformers Fundamentals

Course: AI103
Time Limit: 30 minutes

**Name: _________________________ Date: __________**

## *Instructions:*

• Read each question carefully
• Choose the BEST answer for each question
• Mark your answers clearly

**1. What is the primary advantage of the transformer model over traditional RNNs and LSTMs?**

A. Ability to handle short-range dependencies
B. Ability to handle long-range dependencies
C. Ability to handle only sequential data
D. Ability to handle only non-sequential data

**2. What is the role of the encoder in the transformer model?**

A. To generate an output sequence of symbols one element at a time
B. To map an input sequence of symbol representations to a sequence of continuous representations
C. To perform multi-head self-attention mechanism
D. To perform simple, position-wise fully connected feed-forward network

**3. What is the mathematical representation of the transformer model based on?**

A. Scaled dot-product attention mechanism
B. Simple, position-wise fully connected feed-forward network
C. Multi-head self-attention mechanism
D. Recurrent neural network (RNN) architecture

**4. What is the primary difference between the transformer model and traditional RNNs and LSTMs?**

A. Transformer model uses recurrent connections, while RNNs and LSTMs use feed-forward connections
B. Transformer model uses feed-forward connections, while RNNs and LSTMs use recurrent connections
C. Transformer model uses multi-head self-attention mechanism, while RNNs and LSTMs use single-head self-attention mechanism
D. Transformer model uses simple, position-wise fully connected feed-forward network, while RNNs and LSTMs use complex, position-wise fully connected feed-forward network

**5. What is one of the applications of the transformer model in natural language processing?**

A. Image classification
B. Speech recognition

C. Machine translation
D. Game playing

**6. What is the role of the multi-head attention mechanism in the transformer model?**

A. To jointly attend to information from different representation subspaces at different positions
B. To perform simple, position-wise fully connected feed-forward network
C. To generate an output sequence of symbols one element at a time
D. To map an input sequence of symbol representations to a sequence of continuous representations

**7. What is the primary advantage of the transformer model in terms of parallelization?**

A. Transformer model is not parallelizable
B. Transformer model is partially parallelizable
C. Transformer model is highly parallelizable
D. Transformer model is only parallelizable for sequential data

**8. What is the common misconception about the transformer model?**

A. Transformers are only used in natural language processing
B. Transformers are not parallelizable
C. Transformers are not suitable for long-range dependencies
D. All of the above

**9. What is the primary difference between the transformer model and traditional RNNs and LSTMs in terms of handling long-range dependencies?**

A. Transformer model is not suitable for long-range dependencies, while RNNs and LSTMs are
B. Transformer model is suitable for long-range dependencies, while RNNs and LSTMs are not
C. Transformer model and RNNs and LSTMs are equally suitable for long-range dependencies
D. Transformer model is only suitable for short-range dependencies

**10. What is one of the applications of the transformer model in computer vision?**

A. Speech recognition
B. Machine translation
C. Image classification
D. Game playing

**11. What is the role of the decoder in the transformer model?**

A. To map an input sequence of symbol representations to a sequence of continuous representations
B. To generate an output sequence of symbols one element at a time
C. To perform multi-head self-attention mechanism
D. To perform simple, position-wise fully connected feed-forward network

**12. What is the mathematical representation of the scaled dot-product attention mechanism?**

A. Computes the attention weights as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key
B. Computes the attention weights as a simple, position-wise fully connected feed-forward network
C. Computes the attention weights as a multi-head self-attention mechanism
D. Computes the attention weights as a recurrent neural network (RNN) architecture

**13. What is one of the limitations of traditional RNNs and LSTMs?**

A. Ability to handle long-range dependencies
B. Ability to parallelize the attention mechanism
C. Difficulty in handling long-range dependencies
D. Difficulty in parallelizing the attention mechanism

**14. What is the primary advantage of the transformer model in terms of interpretability?**

A. Transformer model is not interpretable
B. Transformer model is partially interpretable
C. Transformer model is highly interpretable
D. Transformer model is only interpretable for sequential data

**15. What is the role of the multi-head attention mechanism in the transformer model in terms of jointly attending to information?**

A. To jointly attend to information from different representation subspaces at different positions
B. To jointly attend to information from same representation subspaces at different positions
C. To jointly attend to information from different representation subspaces at same positions
D. To jointly attend to information from same representation subspaces at same positions

**16. What is one of the applications of the transformer model in reinforcement learning?**

A. Image classification
B. Speech recognition
C. Game playing
D. Machine translation

**17. What is the primary difference between the transformer model and traditional RNNs and LSTMs in terms of architecture?**

A. Transformer model uses recurrent connections, while RNNs and LSTMs use feed-forward connections
B. Transformer model uses feed-forward connections, while RNNs and LSTMs use recurrent connections
C. Transformer model uses multi-head self-attention mechanism, while RNNs and LSTMs use single-head self-attention mechanism

D. Transformer model uses simple, position-wise fully connected feed-forward network, while RNNs and LSTMs use complex, position-wise fully connected feed-forward network

**18. What is the role of the scaled dot-product attention mechanism in the transformer model?**

A. To compute the attention weights as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key
B. To compute the attention weights as a simple, position-wise fully connected feed-forward network
C. To compute the attention weights as a multi-head self-attention mechanism
D. To compute the attention weights as a recurrent neural network (RNN) architecture

**19. What is one of the common misconceptions about the transformer model?**

A. Transformers are only used in natural language processing
B. Transformers are not parallelizable
C. Transformers are not suitable for long-range dependencies
D. All of the above

**20. What is the primary advantage of the transformer model in terms of handling sequential data?**

A. Transformer model is not suitable for sequential data
B. Transformer model is partially suitable for sequential data
C. Transformer model is highly suitable for sequential data
D. Transformer model is only suitable for non-sequential data

**21. What is the role of the encoder-decoder structure in the transformer model?**

A. To generate an output sequence of symbols one element at a time
B. To map an input sequence of symbol representations to a sequence of continuous representations
C. To perform multi-head self-attention mechanism
D. To perform simple, position-wise fully connected feed-forward network

**22. What is one of the applications of the transformer model in audio processing?**

A. Image classification
B. Speech recognition
C. Machine translation
D. Game playing